

## Q1. Binary Step Function (5 Marks)

The Binary Step Function is an activation function used in single-layer neural networks and perceptron models. It converts continuous input values into binary outputs based on a fixed threshold.

**Equation:**

$$f(x) = \begin{cases} 0, & \text{if } x < 0 \\ 1, & \text{if } x \geq 0 \end{cases}$$

**Explanation:**

- The function is non-differentiable at  $x = 0$  and has zero gradient elsewhere.
- Due to zero gradient, backpropagation cannot update weights effectively.
- Hence, it is not suitable for deep neural networks.
- It can only solve linearly separable binary classification problems.

| Component                  | Marks |
|----------------------------|-------|
| Correct definition         | 1     |
| Mathematical equation      | 1     |
| Gradient property          | 1     |
| Backpropagation limitation | 1     |
| Application / limitation   | 1     |

## Q2. Activation Functions (6 Marks)

### (a) Rectified Linear Unit (ReLU) – 3 Marks

ReLU is the most widely used activation function in deep learning models. It introduces non-linearity while maintaining computational efficiency.

**Equation:**  $f(x) = \max(0, x)$

- Computationally efficient and simple.
- Reduces vanishing gradient problem.
- Allows sparse activation of neurons.
- Limitation: neurons may become inactive (dead neuron problem).

| Component               | Marks |
|-------------------------|-------|
| Definition              | 1     |
| Correct equation        | 1     |
| Advantages / limitation | 1     |

### (b) Leaky ReLU – 3 Marks

Leaky ReLU is a modified version of ReLU designed to overcome the dead neuron problem by allowing a small gradient when the input is negative.

**Equation:**  $f(x) = \alpha x$  for  $x < 0$ , and  $f(x) = x$  for  $x \geq 0$ , where  $\alpha$  is a small constant.

- Maintains non-zero gradient for negative inputs.
- Improves learning stability compared to ReLU.

| Component        | Marks |
|------------------|-------|
| Definition       | 1     |
| Correct equation | 1     |

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|---------------------|
| Advantage over ReLU |
|---------------------|

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|---|
| 1 |
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### Q3. Microprocessor vs Microcontroller (4 Marks)

**Microprocessor:**

- Central processing unit of a computer system.
- Requires external memory and I/O peripherals.
- High processing power and cost.
- Typically follows Von Neumann architecture.

**Microcontroller:**

- Core component of an embedded system.
- Includes CPU, memory, and I/O on a single chip.
- Low power consumption and cost-effective.
- Generally follows Harvard architecture.

| Comparison Aspect        | Marks |
|--------------------------|-------|
| System usage             | 1     |
| Memory & I/O integration | 1     |
| Cost & power consumption | 1     |
| Architecture             | 1     |